PCB Inspection Report

Timestamp	2026-04-04 16:03:34
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

PCB Reliability Engineering Analysis: Missing Hole Defects
1) Root Cause Analysis
The pervasive presence of seven "Missing_hole" defects across multiple locations on the PCB, with five instances exhibiting high confidence scores (0.7954, 0.6899, 0.6693, 0.649, 0.5945), strongly indicates a systemic failure within the PCB fabrication drilling process. The primary contributors are likely:
* **CAM Programming/Data Transfer Error:** The most probable cause is an error in the Computer-Aided Manufacturing (CAM) data generation or transfer from the design files (e.g., Gerber X2, IPC-2581) to the drill machine. Specific drill operations for these holes may have been omitted, misinterpreted, or incorrectly assigned, leading to their complete absence.
* **Drill Tooling Malfunction/Wear:** While less likely to affect seven distinct, distributed locations unless a tool failed immediately after replacement, worn or broken drill bits could cause holes to be partially formed or missed entirely. However, given the "missing" classification, a full absence points away from partial drilling.
* **Drilling Machine Calibration/Maintenance:** Malfunctions in the drilling machine's XYZ axis calibration, spindle control, or depth control could lead to specific drill cycles being skipped or performed incorrectly.
* **Panelization/Fiducial Registration Issues:** If these missing holes are related to tooling or registration features, an error in panelization or the setup of the drilling machine's registration system could contribute, though the variety of BBOX locations suggests more than just a single registration offset.
2) Operational Risks
The presence of multiple "Missing_hole" defects, especially those with high confidence, presents severe operational risks across the product lifecycle:

* **Assembly Impossibility/Line Stoppage:** If these holes are designated for through-hole component leads (THT), mechanical fasteners, or alignment pins, component insertion and subsequent soldering will be impossible. This will lead to immediate assembly line stoppages, significant rework, or outright scrap of assembled units. The high number of defects (7 instances) implies widespread assembly failure.
* **Immediate Electrical Failure:** If any of these missing holes are critical vias (e.g., power, ground, signal integrity paths), the board will exhibit open circuits, short circuits, or incorrect functionality immediately upon power-up and during In-Circuit Test (ICT) or Functional Test (FCT). This directly contributes to the 70.06% failure probability.
* **Compromised Mechanical Integrity:** If these are mounting holes for the PCB itself or for heavy components, their absence will prevent proper securement, leading to board vibration, component detachment, and premature mechanical failure in the field.
* **Thermal Management Issues:** In cases where holes are intended for heatsink attachment, their absence compromises the thermal pathway, leading to localized overheating and accelerated component degradation.
3) Failure Prediction
Based on the "DEFECTIVE" status and "HIGH" risk level with a 70.06% failure probability, failure prediction is immediate and absolute:
* **Pre-Assembly/Pre-Test Rejection:** PCBs containing these "Missing_hole" defects, particularly the five high-confidence instances (0.7954 to 0.5945), will be rejected at the first inspection point (e.g., incoming PCB inspection, automated optical inspection (AOI) at assembly house) or will fail immediately during the component placement phase.
* **Guaranteed Functional Failure:** Should a board somehow bypass initial inspections and proceed to assembly, the absence of critical electrical vias or component through-holes will result in guaranteed functional failure during initial power-on or at the first functional test station (ICT/FCT).
* **No Field Survivability:** There is no plausible scenario where a PCB with seven "Missing_hole" defects, especially critical ones, could function reliably in the field. Any attempt to use such a board would result in immediate or very rapid failure, rendering the end-product non-functional.
4) Repair Explanation
Repairing "Missing_hole" defects on a finished PCB is generally not feasible or advisable:
* **Not Repairable for Critical Features:** For critical through-hole vias, plated mounting holes, or component lead holes, manually drilling a missing hole post-fabrication is extremely risky. It introduces the high probability of damaging internal traces, causing delamination, creating uncontrolled debris, and compromising the copper plating and aspect ratio requirements for reliable electrical connection.
* **Cost-Prohibitive and Risky:** The labor, specialized equipment, and extremely high risk of secondary damage associated with attempting to add multiple missing holes (7 instances) far outweigh the cost of scrapping the defective PCBs. The reliability of such a "repaired" board would also be severely compromised.
* **Recommendation: Scrap:** Due to the fundamental nature of the defect and the high number of occurrences, these PCBs should be unequivocally classified as scrap. No attempt at repair should be made, as it would introduce unacceptable risks and compromise product reliability.
5) Engineering Insights
The consistent detection of "Missing_hole" defects across several locations demands immediate and thorough engineering investigation and process corrective actions:
* **Immediate Fabrication Process Audit:** Initiate an urgent audit of the PCB fabricator's drilling process. This must include a detailed review of the CAM programming, drill file generation, drill machine calibration logs, maintenance schedules, and drill bit tooling inventory/lifetime tracking.
* **Design File vs. CAM File Verification:** Perform a rigorous cross-verification of the original PCB design data (Gerber/ODB++) against the actual CAM files used by the fabricator's drilling machine. Ensure no data translation errors are occurring during the handoff.

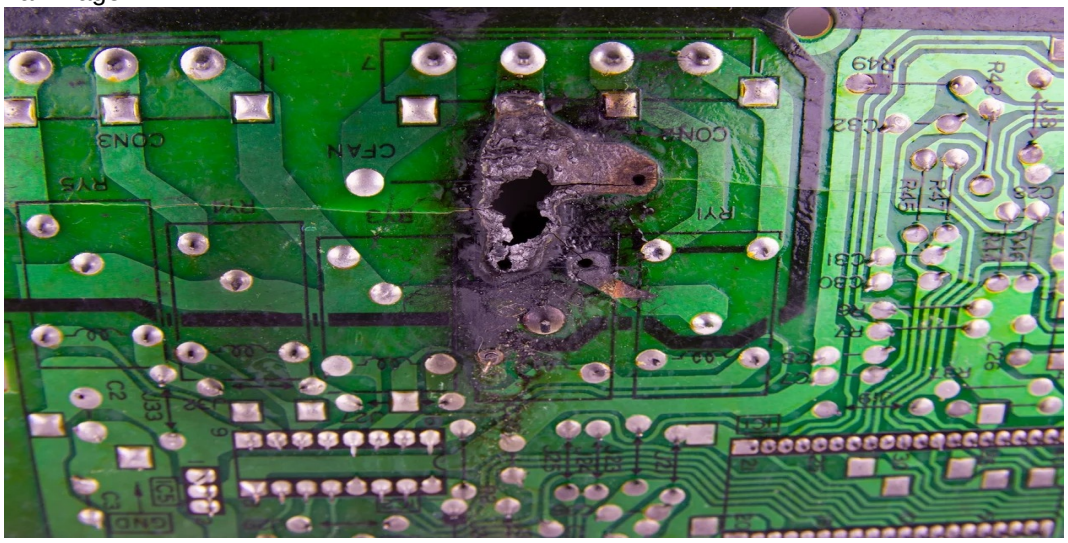
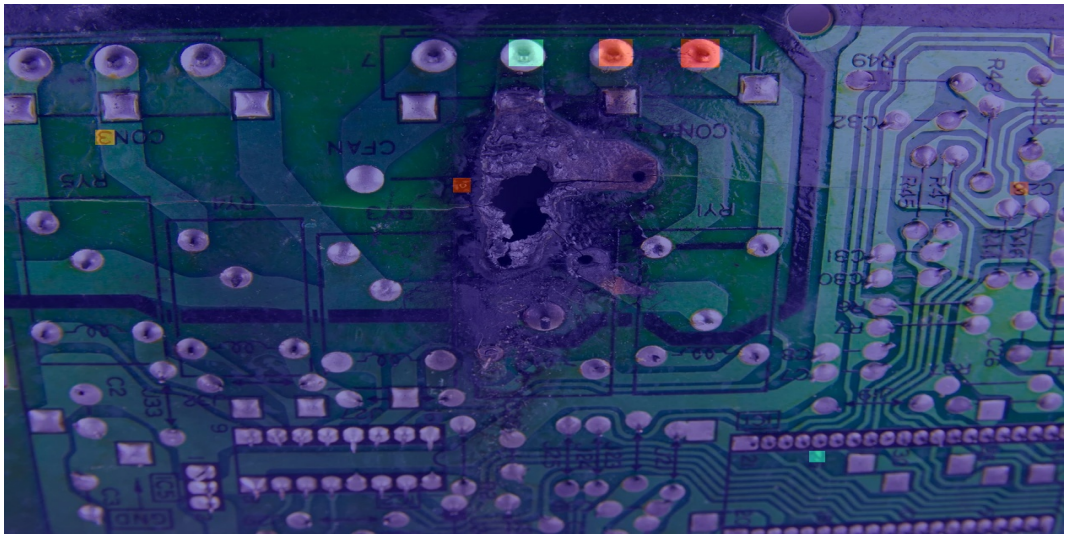
* **Enhanced First Article Inspection (FAI):** Strengthen FAI protocols for all new PCB fabrication runs, with specific emphasis on physical inspection of drill patterns and comprehensive drill report analysis from the fabricator.
* **Automated Optical Inspection (AOI) Calibration:** Verify the calibration and detection algorithms of incoming PCB AOI systems to ensure consistent and accurate detection of missing features, as demonstrated by the current inspection data's high confidence values for these defects.
* **Statistical Process Control (SPC):** Work with the fabricator to implement SPC on their drilling operations to monitor trends in drill size accuracy, positional accuracy, and tool wear, proactively identifying deviations before they lead to such critical defects.
* **Supplier Quality Review:** This type of pervasive, critical defect warrants a formal Supplier Corrective Action Request (SCAR) and potentially an on-site quality audit of the fabricator.

Repair Suggestions

- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Heatmap Overlay